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An Alternative High-Density Ratio Pseudo-potential Lattice Boltzmann Model with Surface Tension Adjustment Capability

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Abstract
In this paper, an alternative model is presented to simultaneously achieve thermodynamic consistency, leading to high-density ratios, and surface tension adjustment capability based on the pseudo-potential lattice Boltzmann model (LBM). These features are achieved by adding two relatively simple terms, each with an adjustable coefficient, selected from the Taylor expansion of the original interaction force, one for achieving and maintaining thermodynamic consistency, and the other for adjusting the surface tension independent of the density ratio. The model also takes advantage of being independent of using SRT, MRT, 2D or 3D lattice models. The capability of the model is evaluated and verified by performing several static and dynamic benchmark test cases. First, the coexistence densities extracted for a flat interface from the model are compared with those of the Maxwell equal area rule in a wide temperature range. Results show that the thermodynamic consistency is well achieved and the coexistence densities are independent of the surface tension strength as well. Next, the well-known Laplace law for a droplet is evaluated and satisfied with the model. In addition, a wide range of surface tension values is achievable at a fixed temperature by adjusting the surface tension adjustment coefficient. After that, the ellipsoidal droplet oscillation is simulated with a good agreement between the analytical and numerical results for the oscillation period. The spurious velocities around the droplet are then evaluated and shown to be reasonable, and comparatively low even at a relatively high-density ratio around of 2400. It was found that the magnitude of these velocities can be further reduced, especially at high-density ratios, by reducing the liquid to gas dynamic viscosity ratio or equivalently increasing the gas to liquid kinematic viscosity ratio. Next, the stability, and the fascinating phenomenon of symmetric bifurcation of a rotating planar droplet are simulated and formation of two- and four-lobed shapes along with rigid rotation is presented depending on the value of the surface tension. Finally, the droplet impact on a liquid film is simulated at a relatively high-density ratio around of 720, and Reynolds number of 1440. However, the upper limit of physical parameters for simulating this phenomenon is found to be a density ratio of 720 and *Re* and *We* numbers of 6600 and 275, respectively. Effects of the surface tension strength are shown as well. The crown spread radius from the model agrees well with the analytical solution reported in the literature.

Keywords Multi-phase · Pseudo-potential LBM · Surface tension adjustment · High-density ratio · Rotating droplet

Extended author information available on the last page of the article

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1 Introduction

The pseudo-potential LBM has attracted the attention of many researchers and engineers with different specialties to use the method for simulating multi-phase and multi-component flows due to its versatility, simplicity and computational efficiency [1, 2]. The idea of the pseudo-potential multi-phase model is based on the mesoscopic mathematical modeling of the interaction forces existing among molecules of a real fluid, which leads to the phase separation of the fluid at temperatures less than its critical temperature. The interaction potential, which is a function of the fluid density, is introduced in the model to calculate the interaction force exerted on each fluid particle in an almost simple straightforward way. However, it was discovered that the pseudo-potential model has some serious problems and limitations including numerical instability at density ratios higher than 10, and low viscosities; lack of the capability to adjust the surface tension independent of the density ratio; thermodynamic inconsistency; and high spurious velocities around the interface [1, 2]. Therefore, considerable efforts have been conducting to overcome or alleviate these problems leading to many research papers and several books. It is recommended to read the [1–3] references to become more familiar with the limitations, developments, and applications of the pseudo-potential model.

Researches have revealed that the stability and accuracy of the pseudo-potential multi-phase model is mainly influenced by the following important items: the force scheme, which determines how the interaction force is included in the LB equation [4, 5]; mathematical definition of the interaction potential function [6]; the method for calculating the interaction force itself [6]; usage of a single or multi-relaxation time (MRT) collision operator [7]; and order of the interaction isotropy [8].

Among the different force schemes reported in the literature, the Guo et al. [9] force scheme fully recovers the Navier–Stokes equations (NSE) without any additional terms [9–12]. Other schemes such as the exact difference method (EDM) and velocity shift scheme, add extra terms to the NSE which is not desired, although they are more stable than the Guo force scheme when the original interaction force is used [11]. It should be mentioned that, by the word ‘original’ interaction force or model, the authors mean the original Shan–Chen interaction force or model. In addition, ‘original pressure tensor’ is meant the pressure tensor obtained as a result of using the original interaction force, and the Guo force scheme.

Towards the way of achieving high-density ratios in the pseudo-potential multi-phase model, Li et al. [11] found that the instability of the Guo force scheme is due to the thermodynamic inconsistency of the original model, meaning that the coexistence densities from the model are different from those of the Maxwell construction rule in thermodynamics. Because the thermodynamic inconsistency mainly leads to deviation of the gas phase towards much lower values compared to the thermodynamic value and, therefore, the density gradient and as a result, the spurious velocities are increased sharply at the interface leading to the model instability. They achieved thermodynamic consistency by adding a new term to the physical macroscopic velocity obtained by Guo et al. [9]. Static benchmark test cases were evaluated in their modified model using the single relaxation time (SRT) collision operator. Later on [12], they used the MRT collision operator to increase the numerical stability and could simulate the droplet impact on a liquid film, successfully. Some applications of this model are reported elsewhere [13–16]. Nevertheless, the added term to the physical velocity in their SRT model was dependent on only one relaxation time, but in the MRT model, there are more than one relaxation times. As a result, they derived the forcing term in the momentum space mathematically with some further

modifications and ignoring some added terms. In addition, the macroscopic equations must be computed again to obtain the added term to the NSE as well. Their 2D MRT model later was extended to the 3D lattice models such as in [17, 18]. Khajepor et al. [19] also proposed multipseudopotential interparticle interaction (MPI) model which is fully consistent with thermodynamics. Huang and Wu [20] proposed another model by adding a source term to the 2D-MRT LB equation based on the third-order Chapman–Enskog analysis.

As an alternative method in this paper, the original interaction force is modified purposefully based on the desired and predicted change in the pressure tensor such that thermodynamic consistency is achieved for a wide temperature range, instead of modifying the physical velocity or adding a source term. Therefore, the procedure for using this alternative method is independent of using SRT or MRT, and 2D or 3D lattice models. Furthermore, the added term to the NSE is exactly equal to the added term to the interaction force due to using the Guo force scheme without any modification on it. Because as reported in the literature [9–12], no extra terms are added to the NSE using this force scheme. The MRT collision operator is selected in this paper to further increase the numerical stability, and reduce the spurious velocities.

In terms of adjusting the surface tension in the pseudo-potential model, Sbragaglia et al. [8] proposed the multi-range pseudo-potential model in which the nearest and also the next-nearest neighbors are considered for computing the interaction force. In this model, the interaction force and as a result the pressure tensor is controlled by two independent parameters, therefore, the authors argued that the equation of states (EOS) and surface tension can be tuned independently. Based on the multi-range model, Falcucci et al. [21] and Chibbaro et al. [22] proposed new models implementing 8th order isotropy (24 neighbors) for calculation of the interaction force. However, in the work of Huang et al. [4], it was discovered that the density ratio changes considerably with the surface tension adjustment in the multi-range model when the Carnahan-Starling EOS was implemented based on the Yuan and Schaefer method [23]. In other words, when a real EOS is implemented, the two parameters that control the interaction force in the multi-range model do not have independent control on the EOS of the model and the surface tension term. Li and Luo [24] added a source term to the MRT LB equation using the D2Q9 lattice and could successfully adjust the surface tension almost independent of the density ratio. The source term was derived based on the original discrete pressure tensor. Based on the Li and Luo model [24], a few other papers are reported in the literature. Xu et al. [18] extended the Li and Luo model [24] to the D3Q15 lattice with rather complex and considerable mathematical derivations. Hu et al. [25] used the Method of Kupershtokh et al. [6] for calculating the interaction force in which an additional parameter controls the interaction force and as a result the pressure tensor. Therefore, the added parameter could be adjusted for achieving thermodynamic consistency. In terms of independent surface tension adjustment, they combined the Kupershtokh et al. [6] model for calculating the interaction force with the model of Li and Luo [24] proposed for adjusting the surface tension using the 2D-MRT version. The main difference between models of Li and Luo [24] and Hu et al. [25] therefore, is using different methods for achieving thermodynamic consistency, and as a result, high-density ratios. Ammar et al. [26] recently extended the model of Hu et al. [25] to the 3D-MRT version. Huang and Wu [20] also proposed a 2D MRT pseudopotential model capable of independent adjustment of the surface tension and coexistence densities. They similarly added a source term to the MRT LB equation but it was derived based on the third-order Chapman-Enskog analysis. They reported that when a third-order analysis is considered, the obtained continuum pressure tensor is accurate while the discrete pressure tensor is accurate only for a specific condition. Lycett-Brown and Luo [27] proposed a new forcing scheme based on the high-order Taylor expansion analysis to adjust the mechanical stability condition towards

achieving high-density ratios. They also could tune the surface tension independent of the density ration by adding a new term similar to the one proposed in [24]. Later, in order to increase the model stability, they developed the LBGK model to the MRT cascaded lattice Boltzmann by applying the third-order Chapman-Enskog analysis [28]. They achieved density ratio up to 1000, Reynolds number 6200 and Weber number 440 in simulating 3D binary droplets collisions. Interested readers can refer to the review paper of Li et al. [29] for more details on the development of the pseudopotential model with covering further papers.

In this paper, as an alternative method, both the thermodynamic consistency (high-density ratios) and independent adjustment of the surface tension are achieved simultaneously by adding one corresponding purposeful term to the original interaction force for each feature. The added terms modify the pressure tensor such that the surface tension and mechanical stability condition can be adjusted independently with a simple consideration. Furthermore, only the nearest-neighbor interactions are considered, which keeps the simplicity of the original model, and the ease of handling the boundary nodes. Another more important and considerable advantage of the current model is its inherent generality due to just modifying the interaction force which makes it independent of using SRT, MRT, 2D or 3D lattice models.

2 Numerical Model

Although the new model proposed in this paper is general in terms of using SRT or MRT LB equations, the MRT model is preferred due to the capability of achieving lower viscosities and higher stability.

2.1 Multi-relaxation-Time LB Model

The MRT LB equation using the Guo force scheme is given by [10]:

$$f_\alpha\left(\boldsymbol{x}+\boldsymbol{e}_\alpha\delta t, t+\delta t\right) = f_\alpha(\boldsymbol{x},t) - \sum_i \Lambda_{\alpha i}\left(f_i - f_i^{eq}\right) + \delta t\left(s_\alpha - \frac{1}{2}\sum_i \Lambda_{\alpha i} s_i\right) \tag{1}$$

where $\boldsymbol{x}$ is the spatial position, $\boldsymbol{e_\alpha}$ the discrete velocity in the αth direction, δt the time step, f the particle distribution function, Λ the relaxation matrix and s is the forcing term in the velocity space. The equilibrium distribution function considering the low Mach limit is given by [1]:

$$f_\alpha^{eq} = \omega_\alpha \rho\left[1 + \frac{\boldsymbol{e}_\alpha\cdot\boldsymbol{V}}{c_s^2} + \frac{\left(\boldsymbol{e}_\alpha\cdot\boldsymbol{V}\right)^2}{2c_s^4} - \frac{\boldsymbol{V}\cdot\boldsymbol{V}}{2c_s^2}\right] \tag{2}$$

The forcing term according to the Guo force scheme is obtained to be [14, 17]:

$$s_\alpha = \omega_\alpha\left[\frac{\boldsymbol{e}_\alpha - \boldsymbol{V}}{c_s^2} + \frac{\left(\boldsymbol{e}_\alpha\cdot\boldsymbol{V}\right)}{c_s^4}\boldsymbol{e}_\alpha\right]\cdot\boldsymbol{F} \tag{3}$$

where ω_α is the weight factor in the αth direction, c_s the speed of sound in lattice, $\boldsymbol{F}$ the total force exerted on each fluid particle, and $\boldsymbol{V}$ the physical macroscopic velocity that is given by:

$$\rho \boldsymbol{V} = \sum_{\alpha} f_\alpha \boldsymbol{e}_\alpha + \frac{\delta t}{2}\boldsymbol{F} \tag{4}$$

where $\rho = \sum f_\alpha$ is the macroscopic density. In this paper, the D2Q9 lattice model is used with grid spacing $\delta x = \delta y = 1$, $c_s^2 = 1/3$, and $\delta t = 1$.

The right-hand-side of Eq. (1) is transformed into the momentum space by multiplying it with the transformation matrix $\mathbf{M}$ as follows [12]:

$$\mathbf{M}(\text{RHS of Eq. (1)}) = \boldsymbol{m} = \hat{f}_i - \sum_i \hat{\Lambda}_{\alpha i}\left(\hat{f}_i - \hat{f}_i^{eq}\right) + \delta t \sum_i \left(I_{\alpha i} - \frac{1}{2}\hat{\Lambda}_{\alpha i}\right)\hat{s}_i \tag{5}$$

where $\hat{f}_i = \mathbf{M} f_i$, $\hat{s}_i = \mathbf{M} s_i$, I is the identity matrix and $\hat{\Lambda}$ is the diagonal relaxation matrix given by [12]:

$$\hat{\Lambda} = \text{diag}\left(s_0, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\right) = \text{diag}\left(\tau_\rho^{-1}, \tau_e^{-1}, \tau_\zeta^{-1}, \tau_j^{-1}, \tau_q^{-1}, \tau_j^{-1}, \tau_q^{-1}, \tau^{-1}, \tau^{-1}\right) \tag{6}$$

The bulk and kinematic viscosities are given by Eqs. (7) and (8), respectively:

$$\xi = \left(\frac{1}{s_1} - 0.5\right)c_s^2 \tag{7}$$

$$\upsilon = (\tau - 0.5)c_s^2 \tag{8}$$

In order to have a different kinematic viscosity for each phase, the following linear interpolation is adopted [30, 31]:

$$\tau = \tau_g + \frac{\rho - \rho_g}{\rho_l - \rho_g}\left(\tau_l - \tau_g\right) \tag{9}$$

where subscripts g and l stand for gas and liquid, respectively. Notice that using the above equation, one can increase the gas to liquid kinematic viscosity ratio in order to decrease the dynamic viscosity ratio. This consideration weakens the spurious currents and therefore increases the stability especially at high-density ratios [12, 14].

The streaming step is carried out in the velocity space as follows:

$$f_\alpha\left(\boldsymbol{x} + \boldsymbol{e}_\alpha \delta t, t + \delta t\right) = \mathbf{M}^{-1}\boldsymbol{m} \tag{10}$$

The macroscopic NSE covered by Eq. (1) in which the Guo force scheme is adopted, are as follows [9–12]:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{V}) &= 0 \\ \frac{\partial}{\partial t}(\rho \boldsymbol{V}) + \nabla \cdot (\rho \boldsymbol{V}\boldsymbol{V}) &= -\nabla P + \nu \nabla \cdot \left[\rho\left(\nabla \boldsymbol{V} + (\nabla \boldsymbol{V})^T\right)\right] + \boldsymbol{F} \end{aligned} \tag{11}$$

In this section, the total force $\boldsymbol{F}$ in Eq. (11) equals only the well-known original interaction force of the pseudo-potential model as follows [12, 25]:

$$\boldsymbol{F} = -G\psi(\boldsymbol{x},t)\left[\sum_{\alpha} w\left(\left|\boldsymbol{e}_{\alpha}\right|^{2}\right)\psi\left(\boldsymbol{x}+\boldsymbol{e}_{\alpha},t\right)\boldsymbol{e}_{\alpha}\right] \tag{12}$$

where G is the interaction strength, and $w(1) = 1/3, w(2) = 1/12$ are the weights in the D2Q9 lattice model. The interaction potential ψ is defined as follows to incorporate a real EOS [11, 23]:

$$\psi = \sqrt{\frac{2\left(P_{EOS} - \rho c_s^2\right)}{Gc^2}} \tag{13}$$

where c is the lattice speed that in this paper is equal to one. The role of G in Eq. (13) is only to keep the square root positive, because it is eliminated in Eq. (12). Although Eq. (13) allows using a real EOS, it leads to the thermodynamic inconsistency meaning that coexistence densities from the model do not match with those of the Maxwell construction rule in thermodynamics [11].

In this paper, the Carnahan-Starling (C-S) EOS is used as follows [1]:

$$P_{EOS} = \rho RT\frac{1 + b\rho/4 + (b\rho/4)^2 - (b\rho/4)^3}{(1 - b\rho/4)^3} - a\rho^2 \tag{14}$$

where $a = 0.4963R^2T_c^2/P_c$ and $b = 0.18727RT_c/P_c$, therefore, the critical temperature is obtained to be $T_c = 0.3773a/(bR)$ where in this paper, $b=4$ and $R=1$. Reducing the parameter a increases the interface thickness, and therefore reduces the spurious currents [14].

2.2 The New Method for Achieving High-Density Ratios

The Taylor expansion of the original interaction force given by Eq. (12) is as follows:

$$\begin{aligned}\boldsymbol{F} =& -Gc^2\left[\psi\nabla\psi + \frac{c^2}{6}\psi\nabla\left(\nabla^2\psi\right) + \ldots\right]\\ =& \frac{-Gc^2}{2}\nabla\psi^2 - \frac{Gc^4}{6}\left[\nabla\left(\psi\nabla^2\psi\right) - \nabla^2\psi\nabla\psi\right] + \ldots\\ =& \frac{-Gc^2}{2}\nabla\psi^2 - \frac{Gc^4}{6}\nabla\left(\psi\nabla^2\psi\right) + \frac{Gc^4}{6}\left[\nabla\cdot(\nabla\psi\nabla\psi) - \frac{1}{2}\nabla|\nabla\psi|^2\right]\end{aligned} \tag{15}$$

where $|\nabla\psi|^2 = \left(\frac{\partial\psi}{\partial x}\right)^2 + \left(\frac{\partial\psi}{\partial y}\right)^2$.

The continuum pressure tensor in the pseudo-potential model is calculated as follows [32]:

$$\nabla\cdot\boldsymbol{P} = \nabla\left(\rho c_s^2\right) - \boldsymbol{F} \tag{16}$$

Therefore, according to Eqs. (15) and (16) the pressure tensor of the original model is simply calculated to be [32]:

$$\boldsymbol{P} = \left(\rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}|\nabla\psi|^2 + \frac{Gc^4}{6}\left(\psi\nabla^2\psi\right)\right)\boldsymbol{I} - \frac{Gc^4}{6}\nabla\psi\nabla\psi \tag{17}$$

The added terms to the ρc_s^2 in the parentheses, model and represent the non-ideal part of the equation of state, the right-most term is related to the surface tension. As a result, the normal pressure for a flat interface in a simplified compact form can be written as [11]:

$$P_n = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[\alpha\left(\frac{d\psi}{dn}\right)^2 + \beta\psi\frac{d^2\psi}{dn^2}\right] \tag{18}$$

where n represents the normal direction of the interface. Making a comparison between Eqs. (17) and (18), one can obtain that $\alpha = -1$ and $\beta = 2$. Notice that the term $-\frac{Gc^4}{6}\nabla\psi\nabla\psi$ in Eq. (17) contains the term $\frac{-2Gc^4}{12}\left(\frac{d\psi}{dn}\right)^2$ which along with the term $\frac{Gc^4}{12}|\nabla\psi|^2$ give $\alpha = -2 + 1 = -1$.

Now by introducing the parameter $\varepsilon = -2\alpha/\beta$ in Eq. (18), doing some mathematical algebra and then integrating it, one can obtain the following equation [12, 25]:

$$\int_{\rho_g}^{\rho_l}\left(p_0 - \rho c_s^2 - \frac{Gc^2}{2}\psi^2\right)\frac{\psi'}{\psi^{1+\varepsilon}}d\rho = 0 \tag{19}$$

where $p_0 = P_n$ is the saturated pressure and $\psi' = d\psi/d\rho$. Equation (19) is named the mechanical stability condition of the pseudo-potential model. On the other hand, the Maxwell construction or Maxwell equal area rule is given as follows [2]:

$$\int_{\rho_g}^{\rho_l}\left(p_0 - P_{EOS}\right)\frac{1}{\rho^2}d\rho = 0 \tag{20}$$

A comparison between Eqs. (19) and (20) reveals that in order to make the two equations the same, two conditions must be met simultaneously; first, the equation of state must be given by $P_{EOS} = \rho c_s^2 + \frac{Gc^2}{2}\psi^2$ and second, the interaction potential must be equal to the density, i.e., $\psi = \rho$. The first condition is met by defining the interaction potential by Eq. (13), which indeed violates the second condition. Therefore, theoretically, the two equations cannot be the same. This discrepancy between the mechanical stability condition and Maxwell construction is called the thermodynamic inconsistency, which is identified to be one of the main reasons for the instability of the pseudo-potential model. However, still there is a way to approach the thermodynamic consistency and as a result, higher stability. The only remaining parameter that can control the mechanical stability condition in Eq. (19) is the parameter ε provided that it could be changed and adjusted explicitly. Finding a solution for making the ε adjustable, the mechanical stability condition can be tuned to the point that the coexistence densities from the model coincide with those of the Maxwell construction. In the following paragraph, the new idea of this paper for adjusting ε is explained.

One of the terms in the Taylor expansion of the original interaction force $\boldsymbol{F}$ given by Eq. (15) is the $-\frac{Gc^4}{12}\nabla\left(|\nabla\psi|^2\right)$. This term is in gradient form and therefore according to Eq. (16) is directly absorbed in the pressure tensor. Furthermore, the absorbed term i.e., $\frac{Gc^4}{12}|\nabla\psi|^2$ produces $\alpha_1 = 1$ in the normal pressure equation. Therefore, in this paper, the new idea is to add the term $+\lambda\frac{Gc^4}{12}\nabla\left(|\nabla\psi|^2\right)$ to the original interaction force, where λ is a free input parameter. In this way according to Eq. (16), $-\lambda\frac{Gc^4}{12}|\nabla\psi|^2$ is absorbed in the pressure tensor and as a result, the coefficient $\alpha_2 = -\lambda$ is generated in the normal pressure as well. Finally, a summation of α_1 and α_2 with the $\alpha_3 = -2$ from the term $-\frac{Gc^4}{6}\nabla\psi\nabla\psi$ gives the total α as follows:

$$\alpha = \alpha_1 + \alpha_2 + \alpha_3 = -1 - \lambda \tag{21}$$

.As a result, the parameter ε is modified as follows:

$$\varepsilon = \frac{-2\alpha}{\beta} = 1 + \lambda \tag{22}$$

Therefore, by varying the input parameter λ, the parameter ε, and as a result, the mechanical stability condition can be adjusted towards the thermodynamic consistency. Thus, the following modification on the original interaction force is proposed in order to achieve thermodynamic consistency, and as a result, high-density ratios:

$$\boldsymbol{F}_{\boldsymbol{new}} = \boldsymbol{F} + \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) \tag{23}$$

According to the Eq. (16) the pressure tensor is modified as follows:

$$\begin{aligned} \nabla \cdot \boldsymbol{P}_{\boldsymbol{new}} &= \nabla \left(\rho c_s^2 \right) - \boldsymbol{F}_{\boldsymbol{new}} \\ \rightarrow \boldsymbol{P}_{\boldsymbol{new}} &= \left(\rho c_s^2 + \frac{Gc^2}{2} \psi^2 + (1-\lambda) \frac{Gc^4}{12} |\nabla \psi|^2 + \frac{Gc^4}{6} \left(\psi \nabla^2 \psi \right) \right) \mathbf{I} - \frac{Gc^4}{6} \nabla \psi \nabla \psi \end{aligned} \tag{24}$$

The first derivatives in Eq. (23) are calculated as follows [30]:

$$\frac{\partial \phi}{\partial r} = \sum_{\alpha} w \left(\left| \boldsymbol{e}_{\alpha} \right|^2 \right) \phi \left(\boldsymbol{x} + \boldsymbol{e}_{\alpha} \right) \boldsymbol{e}_{\alpha(r)} \tag{25}$$

where r is replaced by x or y as 2D Cartesian coordinates to calculate the first derivative in the corresponding direction. The parameter ϕ is first substituted by ψ to calculate the $|\nabla \psi|^2$ and then, after storing this term for each fluid node, is substituted by $|\nabla \psi|^2$ to compute the $\nabla \left(|\nabla \psi|^2 \right)$.

2.3 The New Method for Adjusting the Surface Tension

Similarly, to make the surface tension adjustable, another new term which is $-\kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi$, is added to the original interaction force. The term $-\kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi$ is equal to the two following terms:

$$-\kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi = -\kappa \frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right] \tag{26}$$

On the other hand, the term $+\frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right]$ is present in the Taylor expansion of the original interaction force given by Eq. (15). Therefore, the addition of this term to Eq. (26) leads to the term, $(1-\kappa) \frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right]$. Furthermore, it was mentioned that the surface tension force in the pseudo-potential model is controlled and modeled by the term $\frac{Gc^4}{6} \nabla \cdot (\nabla \psi \nabla \psi)$ in the Taylor expansion of the interaction force or $-\frac{Gc^4}{6} \nabla \psi \nabla \psi$ in the pressure tensor. Therefore, it can be concluded that the new combined term i.e., $(1-\kappa) \frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right]$ is capable of adjusting the surface tension force when the input parameter κ is varied. However, the mechanical stability condition is also influenced by adjusting κ due to the term $(1-\kappa) \frac{Gc^4}{6} \left[-\frac{1}{2} \nabla |\nabla \psi|^2 \right]$ which is not desired. This problem will be discussed later and a simple consideration is applied to resolve it.

Therefore, the final version of the new modified interaction force for achieving thermodynamic consistency as well as surface tension adjustment capability is as follows:

$$\boldsymbol{F}_{\boldsymbol{new}} = \boldsymbol{F} + \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) - \kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi \tag{27}$$

where again it is reminded that $\boldsymbol{F}$ is the well-known original interaction force given by Eq. (12). The final modified pressure tensor is as follows:

$$\begin{aligned} \nabla \cdot \boldsymbol{P}_{\boldsymbol{new}} =& \nabla \left(\rho c_s^2 \right) - \boldsymbol{F}_{\boldsymbol{new}} \\ =& \nabla \left(\rho c_s^2 \right) - \boldsymbol{F} - \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) + \kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi \\ =& \nabla \left(\rho c_s^2 \right) - \boldsymbol{F} - \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) + \kappa \frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right] \end{aligned} \tag{28}$$

$$\rightarrow \boldsymbol{P}_{\boldsymbol{new}} = \left(\rho c_s^2 + \frac{Gc^2}{2} \psi^2 + (1 - \lambda - \kappa) \frac{Gc^4}{12} |\nabla \psi|^2 + \frac{Gc^4}{6} \left(\psi \nabla^2 \psi \right) \right) \boldsymbol{I} - (1 - \kappa) \frac{Gc^4}{6} \nabla \psi \nabla \psi \tag{29}$$

The interaction force $\boldsymbol{F}$ in Eq. (28) is substituted by its Taylor expansion given by Eq. (15). The term $-(1 - \kappa) \frac{Gc^4}{6} \nabla \psi \nabla \psi$ determines the surface tension. However, a special consideration should be taken in order to make the mechanical stability condition independent of the coefficient κ. For this purpose, first, the parameter $\varepsilon = \frac{-2\alpha}{\beta}$ must be computed again for the newly modified pressure tensor. According to Eq. (29) the coefficient α in the normal pressure defined by Eq. (18) is modified as follows:

$$\alpha = (1 - \lambda - \kappa) - 2(1 - \kappa) = -1 - \lambda + \kappa \tag{30}$$

The coefficient $\beta = 2$ has not changed. Therefore, the parameter ε is given by:

$$\varepsilon = \frac{-2\alpha}{\beta} = 1 + \lambda - \kappa \tag{31}$$

Making a comparison between Eqs. (22) and (31) shows that the value $-\kappa$ is added to the ε which means that the mechanical stability condition is undesirably influenced by the κ coefficient. However, the following definition for λ resolves this problem:

$$\lambda = \lambda_0 + \kappa \tag{32}$$

where λ_0 is a tuned value for achieving thermodynamic consistency when $\kappa = 0$. In this way, the parameter ε becomes independent of κ as follows:

$$\varepsilon = 1 + \lambda - \kappa = 1 + \lambda_0 \tag{33}$$

Therefore, in the simulation, first, the tuned value of $\lambda = \lambda_0$ at $\kappa = 0$ is obtained to the point that thermodynamic consistency is appropriately satisfied. Then it is only needed to add the value of κ to the value of λ_0 to preserve the obtained thermodynamic consistency in the model. It should be mentioned that for numerical computation of $\nabla^2 \psi$ in Eq. (27), the following discretization is used which is compatible with the D2Q9 LBM discrete velocity space [30]:

$$\nabla^2 \psi = 2 \sum_{\alpha \neq 0} w \left(|\boldsymbol{e}_\alpha|^2 \right) \left(\psi (\boldsymbol{x} + \boldsymbol{e}_\alpha) - \psi (\boldsymbol{x}) \right) \tag{34}$$

Considering a flat interface along y-direction, the surface tension based on the continuum pressure tensor given by Eq. (29) is obtained as follows:

$$P_n = P_{xx} = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[(-1-\lambda+\kappa)\left(\frac{d\psi}{dx}\right)^2 + 2\psi\frac{d^2\psi}{dx^2}\right] \tag{35a}$$

$$P_\tau = P_{yy} = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[(1-\lambda-\kappa)\left(\frac{d\psi}{dx}\right)^2 + 2\psi\frac{d^2\psi}{dx^2}\right] \tag{35b}$$

$$\sigma_c = \int_{-\infty}^{+\infty}\left(P_{xx}-P_{yy}\right)dx = -\frac{Gc^4}{6}(1-\kappa)\int_{\rho_g}^{\rho_l}\psi'^2\frac{d\rho}{dx}d\rho \tag{36}$$

Mathematical evaluation of the model based on the discrete pressure tensor is presented in the Appendix 1 where it is shown that the mechanical stability condition and surface tension can be adjusted independently based on the discrete pressure tensor as well by applying the same simple consideration of $\lambda = \lambda_0 + \kappa$. The surface tension relation derived from the discrete pressure tensor is obtained to be exactly the same as Eq. (36) obtained from the continuum form. Thus, $\sigma = \sigma_d = \sigma_c = -\frac{Gc^4}{6}(1-\kappa)\int_{\rho_g}^{\rho_l}\psi'^2\frac{d\rho}{dx}d\rho$. Therefore, the model is not influenced by implementing each of the pressure tensors.

3 Results and Discussion

3.1 Evaluation of the Thermodynamic Consistency

To evaluate the thermodynamic consistency of the current model, a liquid layer with a thickness of 50 lattices is placed at the center of a 200×200 domain by means of the following equation [18]:

$$\rho = \rho_g + \frac{\rho_l - \rho_g}{2}\left[\tanh\left(y_1\right) - \tanh\left(y_2\right)\right] \tag{37}$$

where $y_1 = 2(y-75)/5$ and $y_2 = 2(y-125)/5$. The periodic boundary condition is imposed at all sides and the relaxation times are set to $s_0 = s_3 = s_5 = 1$, $s_1 = s_2 = 0.51$, $s_4 = s_6 = 1.1$ and $\tau = 0.6$ for both phases. These relaxation values are kept fixed throughout the rest of the paper unless otherwise mentioned. The coexistence densities from the numerical model after reaching the equilibrium state at $\kappa = 0$ are presented in Table 1 along with those from the Maxwell construction. The parameter a in the EOS is equal to 0.5 unless mentioned in the table.

According to Table 1, it is observed that the stability of the model is increased considerably when λ is changed from zero in the original model to 1.5 in the current modified one. However, values of the vapor density are still far from those of the Maxwell construction. This inconsistency at $\lambda = 1.5$ is increased at lower temperatures which finally leads to the model instability at temperature ratios less than 0.6. Nevertheless, by setting $\lambda = 2.65$,

Table 1 Investigation of thermodynamic consistency of the model at $\kappa=0$

T/T_c	ρ_l/ρ_g		ρ_l/ρ_g	
	Original model ($\lambda=0$)	Current modified model ($\lambda=1.5$)	Current modified model ($\lambda=2.65$)	Maxwell construction
0.8	0.3032/0.0075	0.3057/0.0156	0.3071/0.0211	0.3072/0.0216
0.7	NaN	0.3571/0.0025	0.3582/0.0092	0.3581/0.0093
0.65	NaN	0.3817/0.0004	0.3823/0.006	0.3823/0.0056
0.6	NaN	0.4059/0.0002	0.4062/0.0029 ($a=0.25$)	0.4061/0.003
0.55	NaN	NaN	0.43/0.0015 ($a=0.22$)	0.43/0.0015
0.5	NaN	NaN	0.4541/0.00061 ($a=0.17$)	0.4541/0.00063
0.45	NaN	NaN	0.4786/0.00019 ($a=0.13$)	0.4786/0.0002

the thermodynamic consistency is numerically achieved which in turn greatly enhances the model stability such that density ratios as high as around 2400 at $T/T_c=0.45$ can be achieved. A reduction of parameter a in the EOS at lower temperatures is for keeping the interface thickness fixed in the range of 4-6 lattices.

To verify that the surface tension tuning coefficient κ, does not influence the mechanical stability condition, the coexistence densities at different κ values are also presented in Table 2. As observed, the coexistence densities do not change with κ by applying the simple consideration given in Eq. (32) i.e., $\lambda = \lambda_0 + \kappa$. The coexistence densities are in very good agreements with those of the Maxwell construction.

3.2 Evaluation of the Laplace Law

In this section, the validity of the Laplace law for a droplet i.e., $\Delta p = \sigma/R_0$ is investigated. R_0 is the droplet radius, Δp the pressure difference between the liquid drop and the surrounding gas, and σ the surface tension. A droplet is placed at the center of a 200×200 domain with periodic boundary conditions. Other input parameters are the same as those mentioned in Sect. 3.1 including the tuned value $\lambda_0 = 2.65$. The variation of the pressure difference against the inverse of the drop radius at different values of κ is plotted in Fig. 1. As observed, there is a

Table 2 Model verification on independency of the coexistence densities from the surface tension coefficient κ

T/T_c	ρ_l/ρ_g				
	$\kappa=0$ $(\lambda = \lambda_0 = 2.65)$	$\kappa=-1$ $(\lambda = \lambda_0 + \kappa = 1.65)$	$\kappa=0.6$ $(\lambda = \lambda_0 + \kappa = 3.25)$	$\kappa=0.95$ $(\lambda = \lambda_0 + \kappa = 3.6)$	Maxwell construction
0.8	0.3071/0.0211	0.3071/0.0213	0.3071/0.0211	0.3071/0.021	0.3072/0.0216
0.7	0.3582/0.0092	0.3581/0.0092	0.3581/0.0091	0.3581/0.0091	0.3581/0.0093
0.65	0.3823/0.006	0.3824/0.0059	0.3824/0.006	0.3824/0.006	0.3823/0.0056
0.6	0.4062/0.0029 ($a=0.25$)	0.4062/0.0027 ($a=0.25$)	0.4062/0.003 ($a=0.25$)	0.4062/0.003 ($a=0.25$)	0.4061/0.003

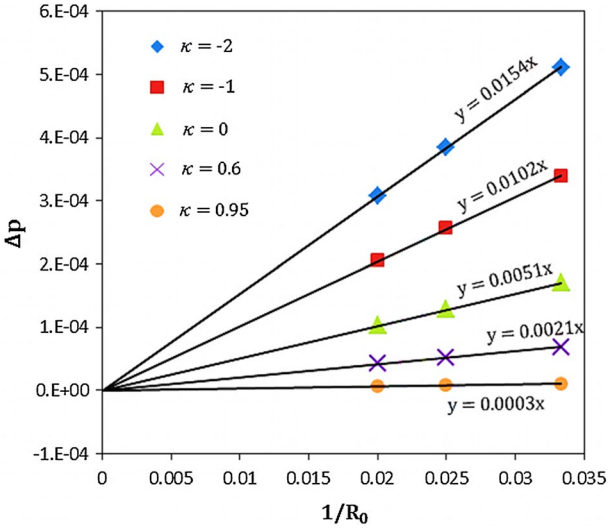


Fig. 1 Evaluation of the Laplace law for a droplet at T=0.6T_c and different κ values

linear relationship between the pressure difference and the inverse of the drop radius at different values of κ. This is an indication that the Laplace law is fully satisfied by the model. Furthermore, the values of the surface tension are calculated and shown in the figure, indicating that a wide range of surface tension values is achievable by varying the value of κ. For example, as shown in Fig. 1, by varying κ in a range of $\kappa \in [-2 : 0.95]$, a wide range of surface tension, $\sigma \in [0.0003 : 0.0154]$ is achieved at T=0.6T_c. This is a manifestation of robustness of the current model for adjusting the surface tension. In addition, the calculated surface tension values in Fig. 1 confirms that the surface tension is proportional to the coefficient $(1-\kappa)$, consistent with Eq. (36). For example, the surface tension value at κ=0 (i.e., 0.0051) is half of the value at κ=−1 (i.e., 0.0102) which is equal to the division of their corresponding $(1-\kappa)$ coefficients, i.e. $\frac{1-0}{1-(-1)} = 0.5$. Notice that the surface tension value is approaching zero by increasing κ to values near one because the coefficient $(1-\kappa)$ is approaching zero.

3.3 Evaluation of the Ellipsoidal Droplet Oscillation

In this section, the ellipsoidal droplet oscillations are investigated at T/T_c=0.45 and for two surface tension values. The computational domain is a 200×200 square with periodic boundary conditions. A vertical ellipsoidal droplet is inserted in the center of the domain using the following equation:

$$\rho = \frac{\rho_l + \rho_g}{2} - \frac{\rho_l - \rho_g}{2}\left\{\tanh\left[\frac{2\left(\sqrt{(x-x_0)^2 + 0.75(y-y_0)^2} - R_0\right)}{5}\right]\right\} \tag{38}$$

where x_0, y_0 are the coordinates of the domain center and R_0=30 lattices. The major radius equals $r_{max} = 34.5$ and the minor one equals $r_{min} = 30$ measured from the center to middle

of the interface. The analytical equation for calculating the oscillation period is given by [14, 33]:

$$T_a = 2\pi \left[n\left(n^2 - 1\right) \frac{\sigma}{\rho_l r_0^3} \right]^{-\frac{1}{2}} \tag{39}$$

where n = 2 is the mode of oscillation and $r_0 = \sqrt{r_{max} r_{min}} = 32.17$ is the droplet radius at its equilibrium state. Variation of r/r_0 against the lattice time is plotted in Fig. 2, where r is the vertical distance between the center and the middle of the interface. Numerical simulation is performed at two surface tension values by setting κ=0 and κ=−1 leading to σ=0.0059 and σ=0.012, respectively. The liquid and gas kinematic viscosity is $\nu_l = 0.01$ and $\nu_g = 0.05$.

The analytical oscillation period is obtained to be 4222 and 2960 for κ=0 and κ=−1, respectively. The corresponding oscillation periods from the simulation results are 4200 and 3000 indicating a good agreement with the analytical values.

3.4 Evaluation of Spurious Velocities

Results of maximum spurious Mach number, $\boldsymbol{M_s^{max}} = \sqrt{u^2 + v^2}/c_s$, around a droplet with radius $R_0 = 30$ are reported in Table 3 at different κ values. u and v are the components of the macroscopic velocity in x and y directions, respectively. In Case (a) of Table 3, the ratio of gas to liquid kinematic viscosities is $\nu_g/\nu_l = 1$ and the values of the viscous relaxation times are $\tau_g = \tau_l = 0.6$. The highest $\boldsymbol{M_s^{max}}$ for this case equals 0.0069 and 0.013 for T=0.6T_c and T=0.45T_c, respectively. Therefore, although the density ratio is highly increased from 140 (T=0.6T_c) to around 2400 (T=0.45T_c), the $\boldsymbol{M_s^{max}}$ still remains small. In Case (b), only the gas to liquid kinematic viscosity ratio, compared to Case (a), is increased from 1 to 5 by increasing τ_g from 0.6 to one. With this increase, the highest $\boldsymbol{M_s^{max}}$ is reduced by 38 percent at T=0.6T_c and 23 percent at T=0.45T_c which is due to a reduction of liquid to gas dynamic viscosity ratio according to $\mu_l/\mu_g = \frac{\rho_l/\rho_g}{\nu_g/\nu_l}$. In order to examine a relatively low liquid viscosity, the viscous relaxation times are reduced from $\tau_l = 0.6$ and $\tau_g = 1$ in case (b) to $\tau_l = 0.52$ and $\tau_g = 0.6$ in case (c).

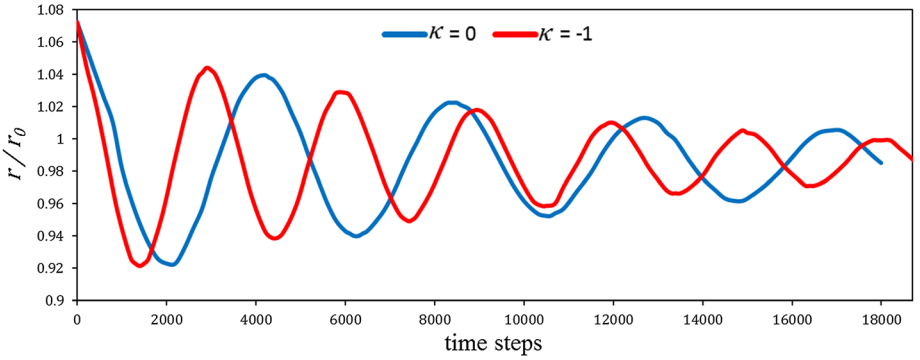


Fig. 2 Ellipsoidal droplet oscillations at T/T_c=0.45 and for two surface tension values

Table 3 Evaluation of spurious velocities

	T/T_c	M_s^{max}			
		$\kappa=-1$	$\kappa=0$	$\kappa=0.6$	$\kappa=0.95$
Case (a)					
$\nu_g/\nu_l = 1$	0.6	0.0069	0.0069	0.0069	0.0068
$\tau_g = \tau_l = 0.6$	0.45	0.013	0.012	0.012	0.013
Case (b)					
$\nu_g/\nu_l = 5$	0.6	0.0043	0.0035	0.0033	0.0031
$\tau_l = 0.6$	0.45	0.01	0.0094	0.0092	0.0094
$\tau_g = 1$					
Case (c)					
$\nu_g/\nu_l = 5$	0.45	0.014	0.012	0.011	0.01
$\tau_l = 0.52$					
$\tau_g = 0.6$					
Case (d)					
$\nu_g/\nu_l = 15$	0.45	0.006	0.007	0.007	0.007
$\tau_l = 0.52$					
$\tau_g = 0.8$					

Furthermore, the temperature in Case (c) is set to T=0.45T_c to evaluate the spurious velocities at a relatively low liquid viscosity, and a very high-density ratio condition. As observed from the results shown in Table 3, the model still remains stable and the highest M_s^{max} equals 0.014. Finally, the kinematic viscosity ratio is increased from 5 in case (c) to 15 in case (d). Similar to the case (b), increasing the kinematic viscosity ratio in case (d) reduced the spurious velocities significantly.

The spurious velocities are shown in Fig. 3 for four test cases reported in Table 3 at κ=0. The length of the vectors is relatively increased in order to better observe patterns of the spurious velocities, their actual magnitude is indeed very low. In Fig. 3a and b, T=0.6T_c and in Fig. 3c and d T=0.45T_c but the kinematic viscosity ratio in Fig. 3(b) and d are higher at the corresponding temperature. As observed from the figure, at a constant temperature, when the kinematic viscosity ratio is increased, the spurious Ma number is decreased and besides the spurious currents move further away from the interface (e.g., compare Fig. 3c and d).

3.5 Simulation of Symmetric Instability of a Rotating Planar Droplet

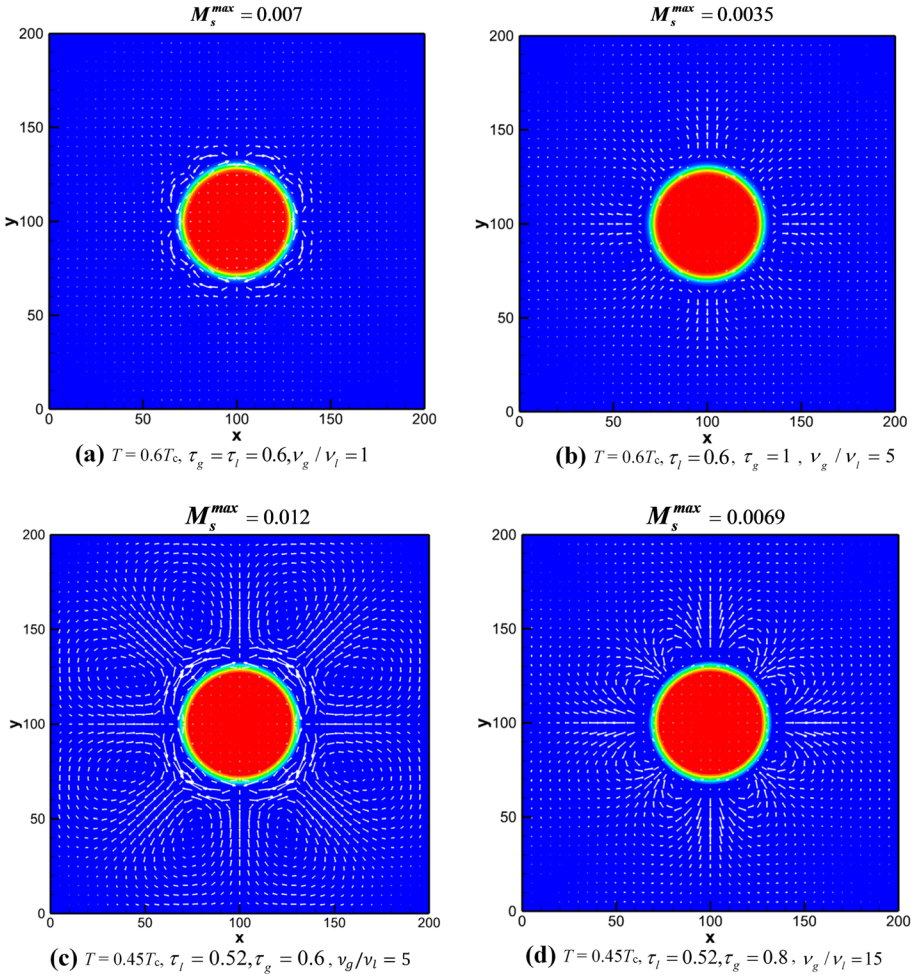
In this section, the instability of a rotating planar droplet with an angular velocity of Ω is investigated. According to Lewis et al. [34] and Marsden [35], for an inviscid planar droplet with radius R_0 and constant surface tension σ, the droplet is stable if the following condition is held:

$$\Omega^2 < 12\sigma/R_0^3 \tag{40}$$

Equation (40) can be equivalently expressed in terms of a condition for a variable surface tension and a constant angular velocity as follows:

$$\sigma > \Omega^2 R_0^3/12 \tag{41}$$

This paper aims to show the surface tension adjustment capability of the proposed model, therefore, Eq. (41) is considered with a constant angular velocity. Therefore, if



(a) $T = 0.6T_c$, $\tau_g = \tau_l = 0.6, \nu_g / \nu_l = 1$

(b) $T = 0.6T_c$, $\tau_l = 0.6$, $\tau_g = 1$, $\nu_g / \nu_l = 5$

(c) $T = 0.45T_c$, $\tau_l = 0.52, \tau_g = 0.6$, $\nu_g / \nu_l = 5$

(d) $T = 0.45T_c$, $\tau_l = 0.52, \tau_g = 0.8$, $\nu_g / \nu_l = 15$

Fig. 3 Spurious velocity vectors around a static droplet at $\kappa = 0$

$\sigma > \Omega^2 R_0^3/12$, then the rotating droplet remains stable. Otherwise, the droplet bifurcates to a two-lobed shape. Furthermore, if $\sigma \leq \Omega^2 R_0^3/60$, then the droplet bifurcates to four branches [34]. In other words, the general condition for the formation of k-lobed shapes is as follows:

$$\sigma \leq \left(\sigma_k = \frac{\Omega^2 R_0^3}{4\left(k^2 - 1\right)} \right) \tag{42}$$

The criterion in Eq. (42) is reported by Lewis et al. (*Remark* 2 of Ref. [34]) derived by implementing the energy-Casimir method [36], a method for studying nonlinear stability in the Hamiltonian systems, and singularity theory. In the current section, the stable rigid rotation, and two- and four-lobed shapes are simulated by adjusting the surface tension strength such that the conditions stated by Eqs. (41) and (42) are met for stable rigid

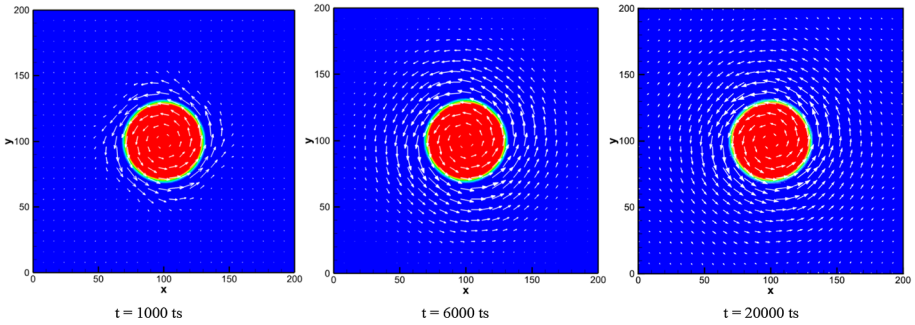


Fig. 4 Stable rotation at $\sigma = 0.028$

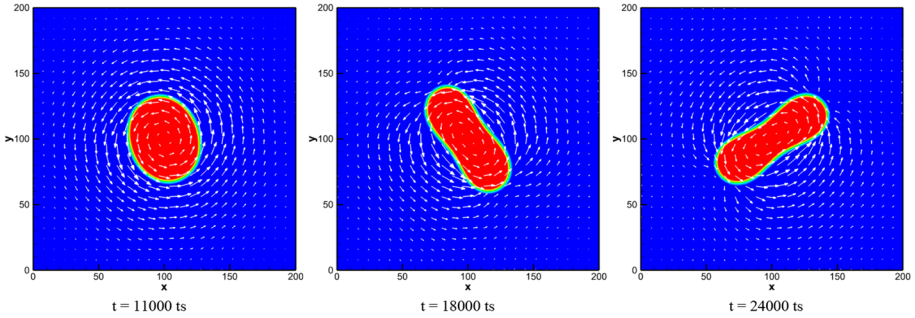


Fig. 5 Two-lobed shape rotation at $\sigma = 0.0176$

rotation, and k-lobed shapes, respectively. Notice that the instability is governed by angular momentum and surface tension and the viscosity effect is negligible.

The simulation domain is a 200×200 rectangle with periodic boundary conditions at all sides. Throughout this section, the density ratio is around 720 corresponding to $T = 0.5T_c$, the gas and liquid kinematic viscosities are $\upsilon_g = 0.033$, and $\upsilon_l = 0.0067$, the droplet radius equals $R_0 = 30$, and the angular velocity is fixed at $\Omega = 0.1/R_0$ all in lattice units. The EOS parameters are $a = 0.25$, $b = 4$, $R = 1$. Therefore, according to Eq. (42), one can simply calculate σ_k values as follows:

$$\sigma_2 = 0.025\,,\ \sigma_3 = 0.0094\,,\ \sigma_4 = 0.005 \tag{43}$$

The simulation is carried out by rotating the droplet by imposing the initial condition $u = -(y - y_0)^*\Omega$ and $v = (x - x_0)^*\Omega$, where x_0 and y_0 are the coordinates of the droplet center. Figure 4 shows results of the rotation at $\sigma = 0.028$ ($\kappa = -3$) being higher than $\sigma_2 = 0.025$, as observed the droplet remains stable.

In Fig. 5, the surface tension is reduced to $\sigma = 0.0176$ ($\kappa = -1.5$) which is less than $\sigma_2 = 0.025$. As observed, the stability is lost and a two-lobed rotating shape is formed. Finally, further reduction of the surface tension strength to $\sigma = 0.004$ ($\kappa = 0.5$) has led to the four-lobed shape as shown in Fig. 6.

It is worth mentioning that attempts for detecting the 3-lobed rotation were failed in our simulations. It was found that the 3-lobed rotation is an unstable mode and moves towards the 2- or the 4-lobed rotations, or leads to irregular shapes without any symmetry.

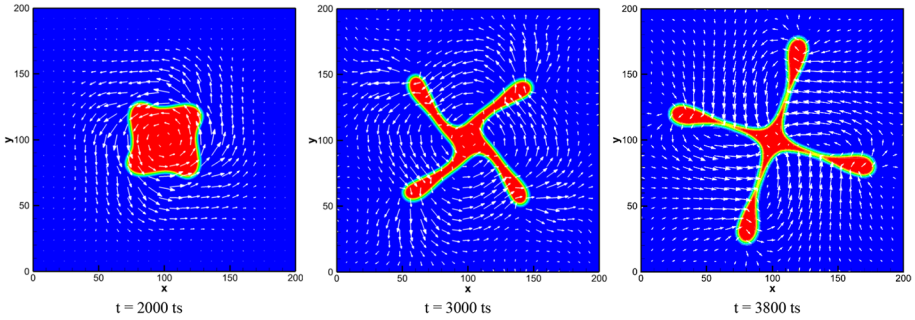


Fig. 6 Four-lobed shape rotation at $\sigma = 0.004$

3.6 Simulation of Droplet Impact on a Liquid Film

The simulation of a droplet impact on a liquid film is a good dynamic benchmark test case to evaluate and observe the model capability for adjusting the surface tension at relatively high-density ratios, and low viscosities. The results of simulation during a droplet impact at $Re = V_0 D/\nu_l = 1440$, $We = \frac{\rho_l V_0^2 D}{\sigma} = 90$, $h_0 = 20$, and $T = 0.5T_c$ is shown in Fig. 7, where $V_0 = 0.12$ is the impact velocity, $D = 80$ the droplet diameter,h_0 the liquid film thickness, and $\sigma = 0.0058$ the surface tension all in lattice units. As seen in simulation images displayed in Fig. 7, the formation of the crown and secondary drops, as well as high-pressure regions in the film at the point of the impact, and velocity vectors pattern indicate that the new model properly simulates the dynamics of the problem.

Figure 8 displays the density contours of the impact for six different values of κ in a range of -2 to 0.7 at the same simulation time ($t^* = 2.25$). With this figure, one can observe and evaluate the new model capability for adjusting the surface tension independent of the density ratio in a rather complex dynamic case of the droplet impact on a liquid film as well. As observed from the figure, the crown height is increased by increasing κ, but the crown thickness is reduced leading to the separation of secondary droplets. As was expected, the number of these secondary drops is increased at higher κ values or higher We numbers.

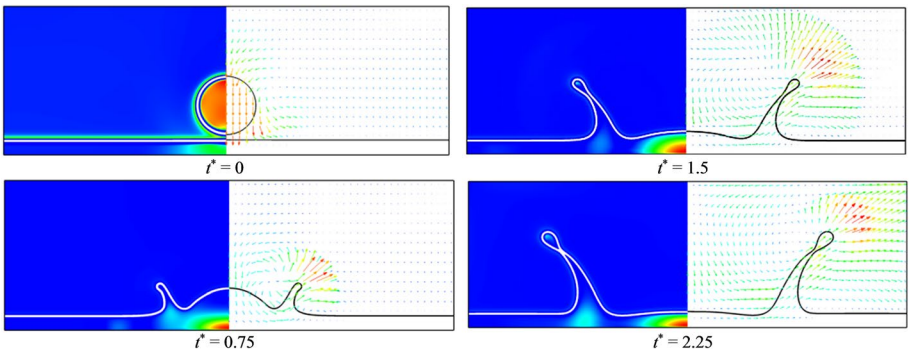


Fig. 7 Pressure contours (left) and velocity vectors (right) during a droplet impact on a liquid film; $T = 0.5T_c$, $Re = 1440$,$We = 90$, $\nu_g/\nu_l = 5$

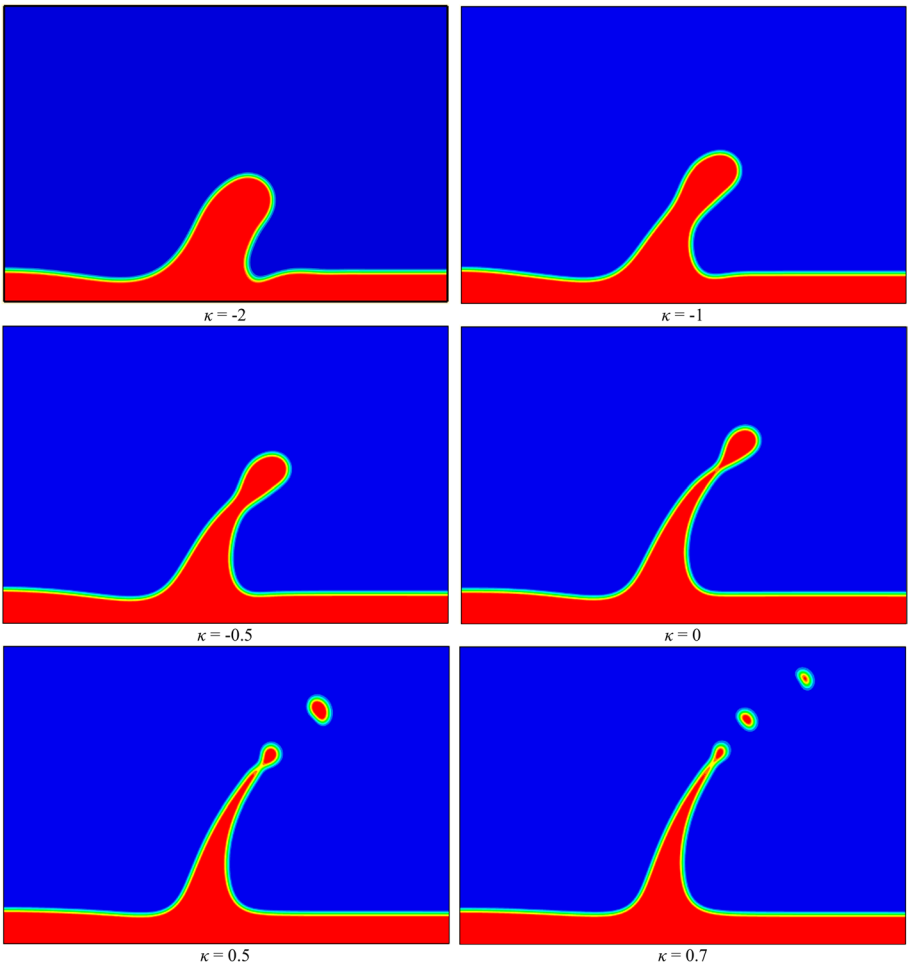


Fig. 8 Effect of surface tension on a drop impact on a liquid film at $t^*=2.25$; $T=0.5T_c$, $h/D=0.25$, $Re=1440$, $\nu_g/\nu_l=5$

Variations of the non-dimensional crown radius (r_c/D) with the dimensionless time ($t^*=tV_0/D$) at two different surface tension or *We* values are plotted in Fig. 9 and the results are compared with the corresponding analytical solution reported by Yarin and Weiss [37].

The analytical solution for the spread radius is given as [37]:

$$r_c/D = \left(\frac{2}{3}\right)^{1/4} \frac{V_0^{1/2}}{D^{1/4}h_0^{1/4}} t^{1/2} \tag{44}$$

Multiplying Eq. (44) by $\frac{D^{1/4}}{D^{1/4}}$ and applying the substitution $t^* = tV_0/D$, one can simply obtain the following equation:

$$r_c/D = \left(\frac{2D}{3h_0}\right)^{1/4} (t^*)^{0.5} \tag{45}$$

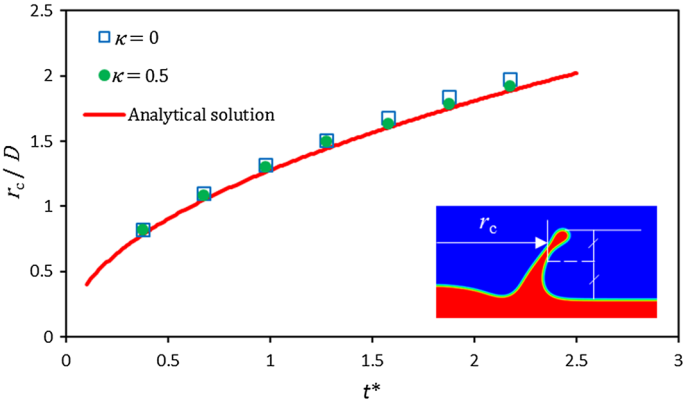


Fig. 9 Crown spread radius compared with the analytical solution of Yarin and Weiss [37]

Substituting for $D=80$ and $h_0=20$ lattices selected for the simulations, the corresponding analytical solution would be:

$$r_c/D = 1.278(t^*)^{0.5} \tag{46}$$

Equation (46) is depicted in Fig. 9 indicating that the simulation results are in good agreement with those of the analytical solution. In addition, the simulation results are almost independent of the *We* number the same as in the analytical solution. Therefore, the spread radius of the crown obeys the so-called power law that was reported in the experiments available in the literature [38] as well.

Critical simulations of the droplet impact on a thin liquid film discovered that the current numerical model is stable for physical parameters as high as $\rho_l/\rho_g = 720$, $\mathrm{Re} = 6600$ and $We = 275$, by setting the following values: $T = 0.5T_c$, $V_0 = 0.11$, $D = 100$, $\nu_l = 0.00167$, $\nu_g = 0.03$, $\sigma = 0.002(\kappa = 0.7)$, $a = 0.13$ (in C - S EOS)

4 Conclusion

In this paper, an alternative method was proposed in the pseudo-potential multi-phase LBM to achieve thermodynamic consistency and, as a result, high-density ratios along with the capability to adjust the surface tension strength independent of the density ratio. Based on the Taylor expansion of the original interaction force, two relatively simple terms were added to it, one for adjusting the mechanical stability condition and the other for adjusting the surface tension. The presented results show that the new model can successfully achieve the thermodynamic consistency. The coexistence densities were obtained to be independent of κ, which adjusts the surface tension strength, indicating that the model is capable of adjusting the surface tension independent of the density ratio. Several benchmark test cases were carried out successfully for the evaluation and verification of the new model. The results for evaluating the Laplace law for a droplet indicate that the surface tension can be adjusted in a wide range. The oscillation period of the oscillating droplet is in a good agreement with the analytical result. The spurious velocities were calculated to be reasonably low in the model even at relatively high-density ratios. The spread radius

of the crown in a drop impact on a liquid film at a relatively high-density ratio and high Re number was found to obey the power law with a good agreement with an analytical solution reported in the literature. Therefore, it is concluded that the new model is capable of simulating multi-phase flows at relatively high-density ratios, and low viscosities along with the capability to adjust the surface tension independent of the density ratio without adding a considerable complexity. The model takes advantage of being general in terms of implementing 2D, 3D, SRT or MRT lattice models.

Appendix 1

In this section, the model is analyzed mathematically based on the discrete pressure tensor proposed by Shan [39]. The discrete pressure tensor of the pseudopotential model using the original interaction force given by Eq. (12), is obtained as follows [12]:

$$\int (\nabla .\, \mathrm{P}) d\Omega = \int \nabla .(\rho c_s^2 \mathrm{I}) d\Omega - \int \mathrm{F} d\Omega \tag{a1}$$

where $d\Omega$ is a finite volume. Using the Gauss integration theorem:

$$\int \mathrm{P}.d\mathrm{A} = \int \rho c_s^2 \mathrm{I}.d\mathrm{A} - \int \mathrm{F} d\Omega \tag{a2}$$

Then in discrete form:

$$\sum \mathrm{P.A} = \sum \rho c_s^2 \mathrm{I.A} - \sum \mathrm{F} \tag{a3}$$

$$\rightarrow \boldsymbol{P}_d = \rho c_s^2 \boldsymbol{I} + \frac{G}{2} \psi(\boldsymbol{x}, t) \left[\sum_{\alpha} w\left(|\boldsymbol{e}_\alpha|^2 \right) \psi\left(\boldsymbol{x} + \boldsymbol{e}_\alpha, t \right) \boldsymbol{e}_\alpha \boldsymbol{e}_\alpha \right] \tag{a4}$$

The Taylor series expansion of the above equation leads to the following expression:

$$\boldsymbol{P_d} = \left(\rho c_s^2 + \frac{Gc^2}{2} \psi^2 + \frac{Gc^4}{12} \psi \nabla^2 \psi \right) \boldsymbol{I} + \frac{Gc^4}{6} \psi \nabla \nabla \psi \tag{a5}$$

The modified interaction force is again presented below:

$$\begin{aligned} \boldsymbol{F_{new}} &= \boldsymbol{F} + \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) - \kappa \frac{Gc^4}{6} \nabla^2 \psi \nabla \psi \\ &= \boldsymbol{F} + \lambda \frac{Gc^4}{12} \nabla \left(|\nabla \psi|^2 \right) - \kappa \frac{Gc^4}{6} \left[\nabla \cdot (\nabla \psi \nabla \psi) - \frac{1}{2} \nabla |\nabla \psi|^2 \right] \end{aligned} \tag{a6}$$

Therefore, the new discrete pressure tensor is simply given by:

$$\rightarrow \boldsymbol{P}_d^{new} = \left(\rho c_s^2 + \frac{Gc^2}{2} \psi^2 + (-\lambda - \kappa) \frac{Gc^4}{12} |\nabla \psi|^2 + \frac{Gc^4}{12} \psi \nabla^2 \psi \right) \boldsymbol{I} + \frac{Gc^4}{6} (\kappa \nabla \psi \nabla \psi + \psi \nabla \nabla \psi) \tag{a7}$$

Notice that the added terms to the original interaction force in Eq. (a6) are in the gradient form just like the well-known $\nabla \left(\rho c_s^2 \right)$ term, thus they are directly absorbed in the discrete pressure tensor as well.

The normal component of the tensor is obtained as follows:

$$P_n^d = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[(-\lambda+\kappa)\left(\frac{d\psi}{dn}\right)^2 + 3\psi\frac{d^2\psi}{dn^2}\right] \tag{a8}$$

Thus, $\varepsilon_d = \frac{-2\alpha}{\beta} = -\frac{2}{3}(-\lambda+\kappa)$.

As observed, the same as the continuum analysis, defining $\lambda = \lambda_0 + \kappa$ leads to:

$$\varepsilon_d = \frac{2}{3}\lambda_0 \tag{a9}$$

Furthermore, considering a flat interface along y direction, the surface tension relation can be obtained as follows:

$$P_n^d = P_{xx}^d = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[(-\lambda+\kappa)\left(\frac{d\psi}{dx}\right)^2 + 3\psi\frac{d^2\psi}{dx^2}\right] \tag{a10}$$

$$P_\tau^d = P_{yy}^d = \rho c_s^2 + \frac{Gc^2}{2}\psi^2 + \frac{Gc^4}{12}\left[(-\lambda-\kappa)\left(\frac{d\psi}{dx}\right)^2 + \psi\frac{d^2\psi}{dx^2}\right] \tag{a11}$$

$$\sigma_d = \int_{-\infty}^{+\infty}\left(P_{xx}-P_{yy}\right)dx = \frac{Gc^4}{6}\int_{-\infty}^{+\infty}\left[\kappa\left(\frac{d\psi}{dx}\right)^2 + \psi\frac{d^2\psi}{dx^2}\right]dx \tag{a12}$$

$$\int_{-\infty}^{+\infty}\psi\frac{d^2\psi}{dx^2}dx = \psi\frac{d\psi}{dx}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}\frac{d\psi}{dx}d\psi \tag{a13}$$

Knowing $\frac{d\psi}{dx}$ is zero at $\pm\infty$ and $\frac{d\psi}{dx} = \frac{d\psi}{d\rho}\frac{d\rho}{dx} = \psi'\frac{d\rho}{dx}$:

$$\rightarrow \sigma_d = -\frac{Gc^4}{6}(1-\kappa)\int_{\rho_g}^{\rho_l}\psi'^2\frac{d\rho}{dx}d\rho \tag{a14}$$

which is exactly the same relation as Eq. (36) derived from the continuum pressure tensor.

Appendix 2

In this appendix, the current model, the multirange potential [8] and the Li and Luo model [24] are further discussed and compared.

The multirange interaction force proposed by Sbragaglia et al. is given by [8]:

$$\boldsymbol{F} = -\psi(\boldsymbol{x},t)\sum_\alpha w\left(\left|\boldsymbol{e}_\alpha\right|^2\right)\left[G_1\psi\left(\boldsymbol{x}+\boldsymbol{e}_\alpha,t\right) + G_2\psi\left(\boldsymbol{x}+2\boldsymbol{e}_\alpha,t\right)\right]\boldsymbol{e}_\alpha \tag{b1}$$

This interaction force was claimed to be able to adjust the EOS through G_1 and the surface tension through G_2, independently. In part III of their paper [24], Li and Luo made an

analysis of the Sbragaglia et al. model to obtain the corresponding relation for the ε parameter and found the following relation (see Eq. (16) in [24]):

$$\varepsilon = \frac{4G_2}{G_1 + 6G_2} \tag{b2}$$

where it is clear that the parameter ε is affected by both G_1 and G_2 coefficients implementing the multirange potential model of Sbragaglia et al. [8]. It should be mentioned that the interaction potential in the Li and Luo analysis of the multirange model is taken to be $\psi = \sqrt{\frac{2(P_{EOS}-\rho c_s^2)}{(G_1+2G_2)c^2}}$. Therefore, Li and Luo showed that the multirange interaction force is not capable of adjusting the EOS and surface tension independently, implementing real EOSs like the Carnahan-Starling.

In terms of achieving thermodynamic consistency, our approach is considerably different from the Li and Luo one. The parameter ε in Li and Luo model is given by [12]:

$$\varepsilon = -2(\alpha + 24G\sigma)/\beta \tag{b3}$$

where σ is a parameter adjusted to numerically achieve the thermodynamic consistency, achieving by applying the following model proposed earlier by Q. Li et al. [11]:

$$\boldsymbol{F}_\alpha = \omega_\alpha \delta t\left(1 - \frac{1}{2\tau}\right)\left[\frac{\boldsymbol{e}_\alpha - \boldsymbol{V}'}{c_s^2} + \frac{\left(\boldsymbol{e}_\alpha \cdot \boldsymbol{V}'\right)}{c_s^4}\boldsymbol{e}_\alpha\right] \cdot \boldsymbol{F} \tag{b4}$$

In other words, they modified the velocity in the Guo force scheme as follows, using the SRT LB equation [11]:

$$\boldsymbol{V}' = \boldsymbol{V} + \sigma \mathbf{F}/\left(\upsilon\psi^2\right) \tag{b5}$$

where $\boldsymbol{V} = \boldsymbol{u} + \mathbf{F}\delta t/2\rho$ with $\rho\boldsymbol{u} = \sum f_\alpha e_\alpha$ and in the denominator, $\nu = (\tau - 0.5)/3$ being the kinematic viscosity. Therefore, our approach for adjusting the mechanical stability condition ε, i.e., adding the term $+\lambda\frac{Gc^4}{12}\nabla\left(|\nabla\psi|^2\right)$ to the original interaction force, is different and novel, although both models add a source term to the LB equation.

In addition, when developed to the MRT version, the Li and Luo model summarized in Eqs (b4) and (b5), is affected by other relaxation times, especially in Eq. (b5), thus some considerations and modifications are carried out on the force term in the momentum space and, as reported in Li and Luo [12], thus mathematical calculations are needed. Whereas, in the current model the source term in the momentum space is computed by a simple code multiplying the force term by the transformation matrix due to no modifications on the Guo force scheme.

In terms of the approach for adjusting the surface tension independently, again the source term should be calculated mathematically in the Li and Luo model, considering the discrete lattice effects on the source term whereas, computing a Laplacian term is the only needed task to make the surface tension adjustable in the current model. As mentioned before, the multirange interaction force by Sbragaglia et al. is based on modifying the interaction force as it is in the current model. Therefore, modification of the interaction force can be viewed at least a different mathematical modeling and approach. The superiority of the current model over the multirange model includes the independent adjustment of the EOS and surface tension, considering only the nearest neighbors interactions and, therefore, easier treatment of boundary conditions.

The value of ε for achieving thermodynamic consistency in the Li and Luo model is indeed a numerical finding as it is in our model, both models use the same interaction potential given by Eq. (13). However, it is not a hard task to find the consistent ε value and is found to be almost independent of the EOSs implemented. Similarly, the parameter λ for achieving thermodynamic consistency in the current model is found by performing several simulations, but when recognized, its value is almost the same implementing other EOSs. For example, we also implemented the Peng–Robinson (P-R) EOS and the consistent λ is found to be in the range of 2.6 to 2.65, which is almost the same value of 2.65 found for the C-S EOS. The consistent σ in the Li and Luo model is reported to be in the range of 0.11–0.12. Xu et al. [18] extended their model to the D3Q15 lattice and $\sigma=0.12$ was reported for both C-S and P-R EOSs. Therefore, in this regard, the obtained thermodynamically consistent value of ε in the current model, like the Li and Luo one, is almost independent of the EOS and thus it is applicable in the general framework of the pseudopotential LBM. The thermodynamically consistent value of ε for the current model is, therefore, $\varepsilon = 1 + \lambda_0 = 1 + 2.65 = 3.65$, almost independent of implementing different equation of states.

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